

obtained from validating whether the LB method satisfies the stability structure [38, 50], but also the convergence proof of the nonlinear problems can be established [30, 29]. Nevertheless, such stability has only been demonstrated for the D2Q9 orthogonal MRT model and how to prove it for many other MRT models is still unclear. To bridge this knowledge gap, Yong et al. [48] put forward an automatic approach for the stability analysis of MRT-LB models and applied it to ten different MRT-LB models [20, 21, 41, 32, 35, 17, 23]. The key step in this approach is to decompose the Jacobian matrix of the collision term in the LB model into the product of a symmetric semi-negative definite matrix and a diagonally positive definite matrix, in particular, this decomposition process can be automatically verified by a simple computer code. With the aid of this work [48], in our current study, we will develop a high-order LB model for (1) and present its explicit stability structure preserving condition.

The remainder of this paper is structured as follows. In Section 2, we present the MRT-LB model for the d -dimensional diagonal-anisotropic diffusion equation. In Section 3, through the direct Taylor expansion, we first deduce the conditions that ensure the present MRT-LB model to be fourth-order consistent with the diagonal-anisotropic diffusion equation, and then present the fourth-order initialization scheme for the MRT-LB model. Thereafter, the condition which guarantees that the MRT-LB model can satisfy the stability structure is provided, and the relationship between the stability structure preserving condition and the L^2 stability of the MRT-LB model is also discussed. Some numerical experiments are carried out in Section 4, and finally some conclusions are summarized in Section 5.

2. The MRT-LB model for the diagonal-anisotropic diffusion equation

In this section, we will present a unified MRT-LB model for the diagonal-anisotropic diffusion equation (1), where the transformation matrix constructed in a natural way and the $DdQ(2d^2 + 1)$ lattice structure are employed. In particular, for the isotropic diffusion equation, another MRT-LB model with the $DdQ(2d + 1)$ lattice structure [fewer discrete velocities than the $DdQ(2d^2 + 1)$ lattice structure] is also provided.

2.1. Spatial and temporal discretization

For the sake of brevity and to facilitate the following analysis, the physical domain Ω is first discretized into a uniform mesh $\mathcal{L}$ with a lattice spacing $\Delta x > 0$, where $\mathcal{L} = \Delta x \mathbb{Z}^d$, and the corresponding lattice node is denoted by $\mathbf{x}_i$. The time is uniformly discretized as $t_n = n\Delta t$, where $\Delta t > 0$ represents the time step. The lattice velocity in the LB method can then be expressed as $c = \Delta x / \Delta t$. For the diagonal-anisotropic diffusion equation (1), the diffusive scaling ($\Delta t \propto \Delta x^2$) is adopted in this work. Without loss of generality, we can also take $\Delta t = \xi \Delta x^2$, where the parameter $\xi \in \mathbb{R}^+$ and $\xi = 0$ when $\Delta x, \Delta t \rightarrow 0$.

2.2. The MRT-LB model

To develop a high-order LB method for the diagonal-anisotropic diffusion equation (1), we here adopt the general MRT-LB model [21, 6, 7] with some additional adjustable relaxation parameters, which can be used to eliminate some certain high-order truncation errors of the LB method, in which the evolution equation can be written as

$$f_k(\mathbf{x} + \mathbf{c}_k \Delta t, t + \Delta t) = f_k(\mathbf{x}, t) - \sum_{i=1}^q \left[\Lambda_{ki} (f_i - f_i^{eq}) \right] (\mathbf{x}, t) + \Delta t \sum_{i=1}^q \left[\left(\mathbf{I} - \frac{\Lambda}{2} \right)_{ki} R_i \right] (\mathbf{x}, t), \quad k \in \llbracket 1, q \rrbracket,^1 \quad (2)$$

where $f_k(\mathbf{x}, t)$, $f_k^{eq}(\mathbf{x}, t)$, and R_k represent the distribution function, equilibrium distribution function, and discrete source term at position $\mathbf{x}$ and time t , respectively. In terms of the present MRT-LB model (2) for (1), we consider the $DdQq$ lattice structure with the number of the discrete velocities $q = (2d^2 + 1)$. To be specific, the velocity set $\mathbf{c}$ of the